

Speech Tech Week 5

1) In the weather and grass related example discussed in the lecture, which of the following entities constitute the Hidden States in the modeling of a Hidden Markov Model?

1 point

- ☒ Weather
- ☐ State of grass
- ☐ Both A. and B.
- ☐ None of the above

observable
seq. is grass
hidden
state is the weather

2) Given M possible observations and N hidden states, there would be $M * N$ entries in the transition probability matrix.

1 point

- ☒ True
- ☐ False

3 weather
[] → 4 states of grass
4x3

1) Based the weather prediction example discussed in the lecture, say there are 6 possible weather conditions and you are given the information about today's weather. Without the help of the Viterbi Algorithm, how many paths would you need to evaluate to obtain the best possible weather forecast for the next 4 days?

1 point

- ☐ 24
- ☐ 4096
- ☒ 1296
- ☐ 10

$$6^4 = 6 \times 6 \times 6 \times 6 = 36 \times 36$$

$$\begin{array}{r} 36 \\ 36 \\ \hline 216 \\ 108 \times \\ \hline 1296 \end{array}$$

2) While using Hidden Markov Models to build speech recognition systems, which of the following statements would be correct?

1 point

- ☐ The observation sequence is made up of phonemes while the hidden states are the acoustic units.
- ☐ The observation sequence is made up of words while the hidden states are the phonemes
- ☐ The observation sequence is made up of words while the hidden states are the acoustic units.
- ☒ The observation sequence is made up of acoustic units while the hidden states are the phonemes.

because phonemes can never be equally observed

1) Given that there are N hidden states to be found and T observations have been made, what is the time complexity of Viterbi Algorithm?

1 point

- ☐ $O(N^T)$
- ☐ $O(T^2 + NT)$
- ☐ $O(T^2 N)$
- ☒ $O(N^2 T)$

Naive $\rightarrow N^T$
Viterbi $\rightarrow N^2 \cdot T$

2) If O is the sequence of observations made, Q is the set of hidden states and λ is the Hidden Markov Model, then which of the following statements is correct?

1 point

- ☐ Viterbi Algorithm chooses a path that maximizes $P(Q|O, \lambda)$
- ☐ Viterbi Algorithm chooses a path that maximizes $P(Q, O|\lambda)$
- ☒ Both A and B
- ☐ None of the above

$$\begin{array}{c} P(Q, O | \lambda) \\ P(Q | O, \lambda) \end{array}$$

1) Given that $O = O_1, O_2, \dots, O_T$ is the sequence of observations and $Q = q_1, q_2, \dots, q_T$ is the sequence of corresponding hidden states. Also, the HMM $\lambda = (A, B, \pi)$, where A is the transition probability matrix, B is the observation probability distribution and π is the initial state probability distribution. Which of the following can be solved in the HMM setting?

1 point

- ☒ What is the probability that a model generated a sequence of observations? That is, $P(O|\lambda) = ?$
- ☐ Estimation of π given B
- ☐ Estimation of B given A
- ☐ All of the above

$$P(O|\lambda) = ?$$

2) Given that $O = O_1, O_2, \dots, O_T$ is the sequence of observations and $Q = q_1, q_2, \dots, q_T$ is the sequence of corresponding hidden states. Also, the HMM $\lambda = (A, B, \pi)$, where A is the transition probability matrix, B is the observation probability distribution and π is the initial state probability distribution. Which of the following is a key estimation problem when using HMM setting?

1 point

- ☐ Estimation of A given π
- ☒ Estimation of Q given O and λ
- ☐ Estimation of A & B
- ☐ All of the Above

1) Which of the following algorithms are used to obtain a soft assignment for the hidden states?

1 point

- ☐ Viterbi Algorithm
- ☒ Forward Algorithm
- ☐ K-means Algorithm
- ☐ None of the Above

→ forward

2) Which of the following algorithms are used to obtain a hard assignment for the hidden states?

1 point

- ☒ Viterbi Algorithm
- ☐ Forward Algorithm
- ☐ Taylor series approximation
- ☐ None of the Above

→ proper classification

1) Which of the following statement about the transition probability matrix is True?

1 point

- ☐ Sum of elements in each column is 1
- ☒ Sum of elements in each row is 1
- ☐ No specific relation as it is just a matrix
- ☐ Both A and B

[] + 1

2) Which of the following statement is factually correct about the Viterbi algorithm?

1 point

- ☒ Given a Hidden Markov Model and an observation sequence, the Viterbi algorithm helps in arriving at the optimal sequence of hidden states.
- ☐ Given an observation sequence and the observation probability distribution, the Viterbi algorithm helps in arriving at the optimal sequence of hidden states.
- ☐ Both A and B.
- ☐ None of the above

3) If q_t represents the state of the Markov model at time $t, t \geq 1$, Which of the following represents the First order Markov Assumption?

1 point

- ☒ $P(q_t = j | q_{t-1} = i, q_{t-2} = k, \dots, q_1 = a) = P(q_t = j | q_{t-1} = i)$
- ☐ $P(q_t = j | q_{t-1} = i, q_{t-2} = k, \dots, q_1 = a) = P(q_t = j | q_{t-1} = i, q_{t-2} = k)$
- ☐ $P(q_t = j | q_{t-1} = i, q_{t-2} = k, \dots, q_1 = a) = P(q_t = j | q_{t-1} = i, q_1 = a)$
- ☐ $P(q_t = j | q_{t-1} = i, q_{t-2} = k, \dots, q_1 = a) = P(q_t = j | q_{t-1} = i)P(q_1 = a)$

→ only depends on past input

4) Given that $O = O_1, O_2, \dots, O_T$ is the sequence of observations and $Q = q_1, q_2, \dots, q_T$ is the sequence of corresponding hidden states. Also, the HMM $\lambda = (A, B, \pi)$, where A is the transition probability matrix, B is the observation probability distribution and π is the initial state probability distribution. Which of the following is a key estimation problem when using HMMs?

1 point

- ☐ Estimation of A
- ☒ Estimation of Q given O and λ
- ☐ Estimation of A given B
- ☐ All of the above

5) Let $b_j(k)$ represent the probability of making an observation k , while in state j . Let a_{ij} represent the transition probability of moving from state i to state j . From the weather prediction example seen the lectures, let Sunny(S), Rainy(R) and Cloudy(C) are the set of states and the grass being Dry(d), Wet(w) and Soggy(s) are the possible observations. If the current state is Cloudy, what is the probability of the next state being Sunny?

1 point

- ☒ a_{CS}
- ☐ $b_C(s)$
- ☐ a_{SC}
- ☐ $a_{SC}b_C(s)$

a_{ij}
 a_{CS}

6) Given that there are N possible hidden states, which among the following are valid initializations for $\pi_i, \forall 1 \leq i \leq N$, where π_i represents the probability of being in state i initially?

1 point

- ☒ $\pi_i = \frac{1}{N}, \forall 1 \leq i \leq N$
- ☐ $\pi_i = 1, \forall 1 \leq i \leq N$
- ☐ $\pi_i = 0.1, \forall 1 \leq i \leq N$
- ☐ None of the Above

→ averaging method

1) Given a HMM with N hidden states and M observation symbols, let $b_j(k)$ represent the probability of observing event j when in state k . Also, let π_j represent the probability of being in state j initially (π_j 's represent the initial state probabilities). Which of the following statements is True? 1 point

- ☐ $\sum_{k=1}^M b_j(k) \geq 1$ and $\sum_{j=1}^N \pi_j \leq 1, \forall 1 \leq j \leq N$.
- ☐ $\sum_{j=1}^N b_j(k) = 1$ and $\sum_{j=1}^N \pi_j = 1, \forall 1 \leq k \leq M$.
- ☒ $\sum_{k=1}^M b_j(k) = 1$ and $\sum_{j=1}^N \pi_j = 1, \forall 1 \leq j \leq N$.
- ☐ $\sum_{j=1}^N b_j(k) \geq 1$ and $\sum_{j=1}^N \pi_j \leq 1, \forall 1 \leq k \leq M$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\sum_{k=1}^M b_j(k) = 1$ and $\sum_{j=1}^N \pi_j = 1, \forall 1 \leq j \leq N$.

2) Which among the following is NOT a parameter to be estimated while building a HMM? 1 point

- ☐ Transition Probabilities
- ☒ Observation Probability Distribution
- ☒ Observation Sequence
- ☐ Initial State probability distribution

No, the answer is incorrect.

Score: 0

Accepted Answers:

Observation Sequence

Sequence is not a parameter of HMM

3) Which of the following is needed to predict the most probable observation sequence, if we are given the sequence of states the HMM is in? 1 point

- ☐ Viterbi Algorithm
- ☐ Transition Probability Matrix
- ☒ Observation Probability Distribution
- ☐ All of the above

Yes, the answer is correct.

Score: 1

Accepted Answers:

Observation Probability Distribution

4) Given that $O = O_1, O_2, \dots, O_T$ is the sequence of observations and $Q = q_1, q_2, \dots, q_T$ is the sequence of corresponding hidden states. Also, the HMM $\lambda = (A, B, \pi)$, where A is the transition probability matrix, B is the observation probability distribution and π is the initial state probability distribution. Which of the following is a key estimation problem when using HMMs? 1 point

- ☒ Estimation of λ given O
- ☐ Estimation of O given π and q_1
- ☐ Estimation of Q given B
- ☒ All of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Estimation of λ given O

5) Let $b_j(k)$ represent the probability of making an observation k , while in state j . Let a_{ij} represent the transition probability of moving from state i to state j . From the weather prediction example seen the lectures, let Sunny(S), Rainy(R) and Cloudy(C) are the set of states and the grass being Dry(d), Wet(w) and Soggy(s) are the possible observations. If the current state is Sunny, what is the probability of the grass being Dry?

1 point

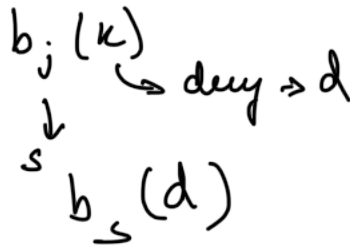
- ☐ $b_S(w)$
- ☐ a_{Sd}
- ☒ $b_S(d)$
- ☐ $a_{SS}b_S(d)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$b_S(d)$



6) Let $b_j(k)$ represent the probability of making an observation k , while in state j . Let a_{ij} represent the transition probability of moving from state i to state j . From the weather prediction example seen the lectures, let Sunny(S), Rainy(R) and Cloudy(C) are the set of states and the grass being Dry(d), Wet(w) and Soggy(s) are the possible observations. The expected number of times for being in the Cloudy state is given by?

1 point

- ☐ $\frac{a_{CC}}{a_{SC}+a_{BC}+a_{CC}}$
- ☐ $\frac{1}{a_{SC}+a_{BC}+a_{CC}}$
- ☒ $\frac{b_C(d)+b_C(w)+b_C(s)}{a_{SC}+a_{BC}+a_{CC}}$
- ☐ $\frac{1}{1-a_{CC}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{b_C(d)+b_C(w)+b_C(s)}{a_{SC}+a_{BC}+a_{CC}}$

7) $M = 3$ be the possible number of observations and $N = 5$ be the number of possible hidden states. We define the HMM $\lambda = (A, B, \pi)$, where A is the transition probability matrix, B is the observation probability distribution and π is the initial state probability distribution. Given the observation sequence, how many parameters are to be estimated to model the HMM?

1 point

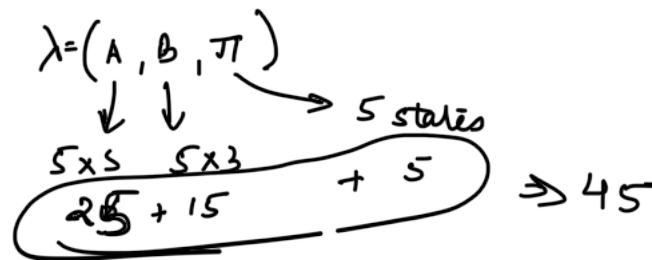
- ☐ 30
- ☐ 15
- ☒ 45
- ☐ 60

Yes, the answer is correct.

Score: 1

Accepted Answers:

45



8) Which of the following is a property of the Transition probability matrix?

1 point

- ☐ Symmetric
- ☒ Stochastic
- ☐ Diagonal
- ☐ Symmetric and Stochastic